KAZAKH BRITISH TECHNICAL UNIVERSITY

WEB DEVELOPMENT

Lesson 7

Python is:

- Dynamic
- Interpreted
- Object-Oriented
- Exceptional
- Comfortable
- Readable
- Community

Interactive Shell

```
$ python
>>> print "Hello, world!"
Hello, world!
>>>
$ python3
>>> print("Hello, world!")
Hello, world!
>>>
```

Comments

```
# Best. Comment. Ever.
```

Booleans and Null

```
True

False

None
```

Strings

- 'Hello, world!’
- "Hello, world!”
- """Hello,
  world!"""
- u"Hëllö, wörld!”

String Operations

```
"foo" + "bar"

"foo"[0]

"foo"[:1]

"foo".upper()

"{0}: {1}".format("foo", "bar")

"{foo}: {bar}".format(foo=42, bar=11)

len(“foo”)
```

String Operations

```
"foo" + "bar" ==> "foobar"

"foo"[0] ==> "f"

"foo"[:1] ==> "f"

"foo".upper() ==> "FOO"

"{0}: {1}".format("foo", "bar") ==> "foo: bar"

"{foo}: {bar}".format(foo=42, bar=11) ==> "42: 11"

len(“foo") ==> "3"
```

Sequence Operation

```
[...][0]

[...][-1]

[...][:1]  # same as [...][0:1]

[...].append(7)

[...].pop()

len([...])
```

Dictionaries

```
{'key1': 'value1', 'key2': 'value2'}
```

Dictionary Operations

```
{...}['key1']

{...}.get('key2')

{...}.keys()

{...}.values()

{...}.items()

len({...})
```

Assignment & Comparison

```
foo = ‘bar’

foo == ‘baz'

foo != ‘baz'

foo is None

foo is not None
```

Flow Control

```
if expression:
    ...
elif expression:
    ...
else:
    ...
```

Flow Control

```
for item in sequence:

    if expression:
        continue
    else:
        break
```

Functions

```
def foo():
    return 42


def foo(bar):
    return bar


def foo(bar, baz=“fit”):
    return (bar, baz)
```

Classes

```
class Foo(object):
   def __init__(self, bar):
      self.bar = bar
```

Docstrings

```
"Modules can have docstrings."

class Foo(object):

    "Classes can have docstrings too."

    def __init__(self, bar):
        self.bar = bar
```

Exceptions

```
try:
    raise Exception("OH NOES!”)

except:
    log_error()
    raise

else:
    do_something_more()

finally:
    clean_up()
```

Namespaces

```
import logging

from datetime import timedelta

from decimal import Decimal as D


from models import Product
```

Style: PEP-8

- Four-space indents
- lower_case_methods
- CamelCaseClasses
- Line breaks around 78-79 chars

Installing Packages

- easy_install: easy_install package
- pip: pip install package

Installing Packages

- Installed packages go into a site-packages directory in your Python lib
- But different programs may need different versions of packages…
- So we have virtual environments!

Virtual Environments

- virtualenv
- Creates an isolated Python environment with its own site-packages
- Install whatever you want without fouling anything else up

Activate the Virtual Environment

```
# Mac/Linux/etc...

$ virtualenv myenv
$ source myenv/bin/activate

# Windows

> python virtualenv myenv
> myenv/Scripts/activate.bat
```

Questions?